

Kayoko Thornton

ket65@cornell.edu • 913-235-0827 • www.linkedin.com/in/kayoko-t
Portfolio: <https://cornell-mae-ug.github.io/fa25-portfolio-KayokoT/>

Education

Cornell University, College of Engineering

Ithaca, NY

Bachelor of Science, Mechanical Engineering. GPA: 4.147/4.30

Aug 2023 – Dec 2026

Relevant Coursework: Mechanics of Engineering Materials, Fluid Mechanics, System Dynamics, Statistics, Waves, Thermodynamics, Statics & Mechanics of Solids

Research Experience

Energy Research Lab

Cornell University

Undergraduate Researcher

Sept 2025 – Present

- Conduct weekly hands-on experiments to optimize a multistage passive cooling system driven by salinity gradients
- Collect and analyze thermocouple data to determine the impact of stage count, saline concentration, and air gap thickness on the cooling performance
- Aim to couple the cooling device with a solar distiller to increase efficiency

Staged Pressurized Oxy-Combustion (SPOC) Lab

Washington University in St. Louis

Thermal Energy Management on Multiple Scales REU Fellow

June 2025 – Aug 2025

- Studied radiative heat transfer and carbon-negative biomass combustion
- Researched and validated a thermal imaging camera for SPOC system monitoring
- Developed a MATLAB script with Modbus TCP protocol for real-time alarm optimization
- Presented at the End-of-Summer Symposium and Poster Palooza

Symbiotic Engineering and Analysis Lab

Cornell University

Undergraduate Researcher

Sept 2024 – May 2025

- Simulated co-located 5 MW wind turbines with various wave energy converters and mooring configurations in WEC-Sim (MATLAB)
- Evaluated performance of wind-wave systems by calculating LCOE and developing data processing scripts
- Validated mooring system recommendations to optimize coupling with wind turbines

Other Experience

ENGRD 2210: Thermodynamics, Dept. of MAE

Cornell University

Undergraduate Teaching Assistant

Sept 2025 – Dec 2025

- Assisted students during weekly discussion sections by presenting course topics, explaining problem-solving strategies, and answering questions
- Managed logistical tasks such as answer key write-ups and test scanning

Plant Polytunnel Design

Cornell Botanic Gardens

Independent Study

Jan 2024 – May 2024

- Researched and drafted designs in AutoCAD for a modifiable, greenhouse-like structure
- Outlined the budget for materials and supplies

Skills

- **Technical:** R, Python, Java, MATLAB, Fusion, AutoCAD, Word, Excel, PowerPoint, Outlook, Slack
- **Language:** French (intermediate); Japanese (intermediate)